

# Sensitivity Comparison of Atomic Interferometers for Gravitational Measurements on Positronium Using Monte Carlo Simulations or Treatise on Shelling Bananas

Mateusz „■” Winiarski

fizyka<sup>†</sup>, FAIS, Uniwersytet Jagielloński

today, May 14, 2026



# contents

what is positronium

why is it interesting

how to do it

why is it hard



# problem



figure: a banana



# problem

| type      | result             | probability |
|-----------|--------------------|-------------|
| $\beta^-$ | $^{40}\text{Ca}$   | 89.6 %      |
| EC        | $^{40}\text{Ar}^*$ | 10.3 %      |
| direct EC | $^{40}\text{Ar}$   | 0.1%        |
| $\beta^+$ | $^{40}\text{Ar}$   | 0.001%      |



figure: a banana, and  $^{40}\text{K}$  decay modes table



# what is a positronium?

$\underbrace{\text{positr}}_{e^+}$   $\underbrace{\text{onium}}_{\text{and } e^- \text{ bound state}}$

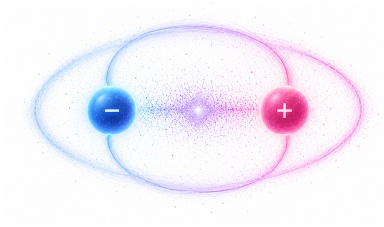


figure: artistic image of positronium



# positronium states

and a quick repeat from atomic physics

- ▶ para-positronium  $1^1S_0$ :  $T_{1/2} = 125$  ps
- ▶ ortho-positronium  $1^3S_1$ :  $T_{1/2} = 142$  ns

what does this mean?

$$n^{2S+1}L_{m_S}$$

$$\vec{J} = \vec{L} + \vec{S}$$



# positronium states

and a quick repeat from atomic physics

- ▶ para-positronium  $1^1S_0$ :  $T_{1/2} = 125$  ps
- ▶ ortho-positronium  $1^3S_1$ :  $T_{1/2} = 142$  ns
- ▶ meta-stable-positronium  $2^3S_1$ :  $T_{1/2} = 1.14$   $\mu$ s

what does this mean?

$$n^{2S+1}L_{m_S}$$

$$\vec{J} = \vec{L} + \vec{S}$$



# positronium states

and a quick repeat from atomic physics

- ▶ para-positronium  $1^1S_0$ :  $T_{1/2} = 125$  ps
- ▶ ortho-positronium  $1^3S_1$ :  $T_{1/2} = 142$  ns
- ▶ meta-stable-positronium  $2^3S_1$ :  $T_{1/2} = 1.14$   $\mu$ s

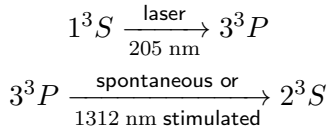
what does this mean?

$$n^{2S+1}L_{m_S}$$

$$\vec{J} = \vec{L} + \vec{S}$$



# how to obtain meta-stable beam?

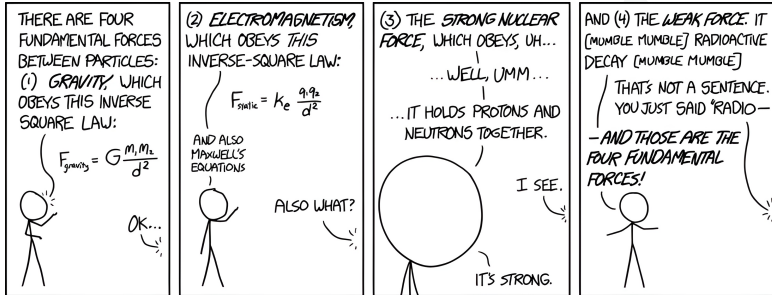


# how to obtain positronium beam?

lasers



# fundamental forces in physics



# fundamental forces in physics

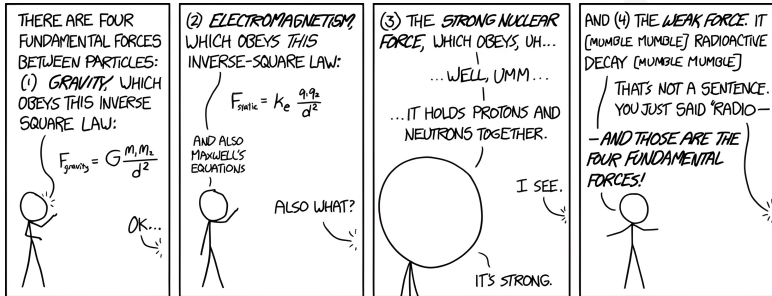


figure: "Of these four forces, there's one we don't really understand."  
"Is it the weak force or the strong—" "It's gravity."



# why do we bother?

*'But in any event, it is in the hands of our generation to perform an experiment to measure the gravitational acceleration of antimatter. Some day it will be done, whether we do it or not. It will be done. If we do not do it, and the answer eventually turns out to be what we expect, then future generations will look back upon us and say it was a shame. But if the answer turns out to be a surprise, then, if we do not do it, future generations will look back upon us and say we were fools.'*

- M. M. Nieto et al. Theoretical Motivation for Gravitation Experiments on UltraLow Energy Antiprotons and Antihydrogen. arXiv, 1994.



## a bit of theory

why are we even bothering to measure it?

- ▶ **CPT symmetry:** according to the CPT theorem, antimatter should experience the same gravitational acceleration as matter
- ▶ **photon behaviour:** photons are their own antiparticles, and they act just like General Relativity laws tell them so
- ▶ **conservation of energy:** if antimatter goes up, we can annihilate it "higher" than it was produced



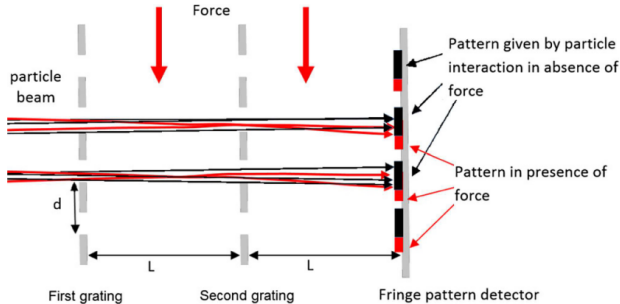
## a bit of theory...

However...

- ▶ **Kowitt theory:** antiparticles are "holes" in Dirac particle sea with negative gravitational mass
- ▶ **Santilli-Villata theory:** antimatter exists in inverted space-time, which is projected to our reality as negative motion
- ▶ **Cabbolet's theory:** particles move in "stepwise" motion, alternating between state of rest and "wave-like" movement
- ▶ **dark matter???**



# Moiré deflectometer

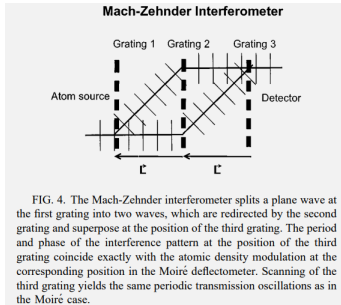


**Fig. 3.** Schematic of a classical two-grating deflectometry system with periodicity  $d$ .

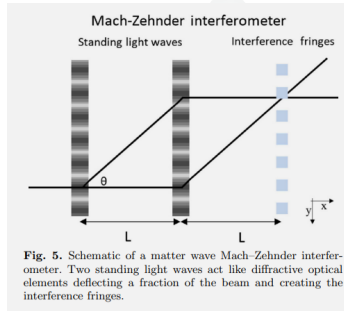
figure: (source: Mariazzi et al., 2020)



# Mach-Zehnder interferometer



(a) (source: Oberthaler et al., 1996)



(b) (source: Mariazzi et al., 2020)



# Talbot-Lau interferometer

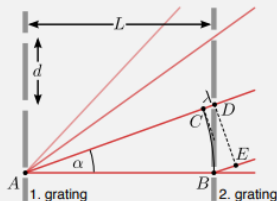


Figure 2.3.: Neighbouring paths in a Talbot-Lau interferometer. Following the argument of Ernst Lau [75], the resulting phase difference between neighbouring slits of the second grating equals the wavelength  $\lambda$  if the grating separation  $L$  corresponds to the half Talbot length. The second grating seems to be illuminated by a plane wave and the intensity field behind the second gratings is similar to the one observed in the Talbot effect.

figure: (source: Bräunig, 2015)



# why is it hard

or: why hasn't it been done before

| device       | 1 $g$ measurement time |
|--------------|------------------------|
| moiré        | 22000 yr               |
| Mach-Zehnder | 5.8 yr                 |
| Talbot-Lau   | 730 yr                 |



# bibliography



Mariazzi, Sebastiano et al. (Apr. 2020). "Toward inertial sensing with a 23S positronium beam". en. In: *The European Physical Journal D* 74.4, p. 79. ISSN: 1434-6060, 1434-6079. DOI: 10.1140/epjd/e2020-100585-8. URL: <https://link.springer.com/10.1140/epjd/e2020-100585-8>.



Oberthaler, Markus K. et al. (Oct. 1996). "Inertial sensing with classical atomic beams". en. In: *Physical Review A* 54.4, pp. 3165–3176. ISSN: 1050-2947, 1094-1622. DOI: 10.1103/PhysRevA.54.3165. URL: <https://link.aps.org/doi/10.1103/PhysRevA.54.3165>.

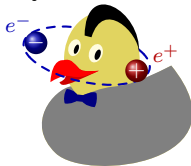


Bräunig, Philippe H. M. (2015). "Atom Optical Tools for Antimatter Experiments". In: DOI: 10.11588/HEIDOK.00018044. URL: <http://archiv.ub.uni-heidelberg.de/volltextserver/id/eprint/18044>.



the end

thank you for attention



<https://github.com/falcotinnunculus/przezrocza/blob/wladca/electricboogaloo/prezka.pdf>



# (backup) why is it hard 2: electric boogaloo

or: why haven't I done it yet

This page is intentionally left blank

